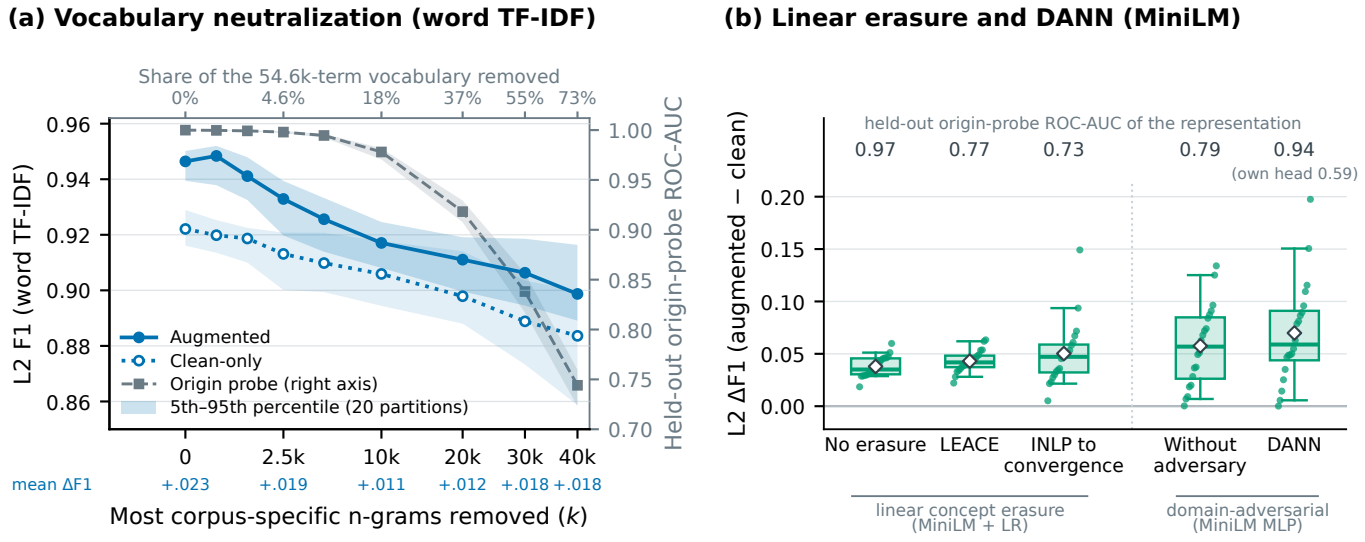
(a) Vocabulary neutralization (word TF-IDF)
Share of the 54.6k-term vocabulary removed
0%
4.6%
18%
37%
55%
73%
L2 F1 (word TF-IDF)
Held-out origin-probe ROC-AUC
0.96
0.94
0.92
0.90
0.88
0.86
1.00
0.95
0.90
0.85
0.80
0.75
0.70
Augmented
Clean-only
Origin probe (right axis)
5th–95th percentile (20 partitions)
0
2.5k
10k
20k
30k
40k
mean ΔF1
+.023
+.019
+.011
+.012
+.018
+.018
Most corpus-specific n-grams removed (k)
(b) Linear erasure and DANN (MiniLM)
held-out origin-probe ROC-AUC of the representation
0.97
0.77
0.73
0.79
0.94
(own head 0.59)
L2 ΔF1 (augmented − clean)
0.20
0.15
0.10
0.05
0.00
No erasure
LEACE
INLP to convergence
Without adversary
DANN
linear concept erasure (MiniLM + LR)
domain-adversarial (MiniLM MLP)